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CO2 AT2

CSA16: Data Warehousing and Data Mining

Question 1: Data Cleaning and Normalization in Healthcare Analytics

A. Missing Value Treatment

i. Missing Values Identified: Age (P2), BMI (P3), Annual_Visits (P5)

ii. Imputation Calculations:

- **Age:** Mean = 48.14 | Median = 42
- **BMI:** Mean = 33.57 | Median = 29.5
- **Annual_Visits:** Mean = 17.29 | Median = 7

iii. Comparison & Justification: Median Imputation is more appropriate than Mean Imputation because the dataset contains extreme values (Age 110, BMI 60.0, Visits 50), which heavily skew the mean upwards. The median remains robust against these extreme values.

B. Outlier Detection

i. Z-Score Method: Calculated using $Z = (X - \mu) / \sigma$.

- **Age:** P4 (Age 110) yields $Z = 2.18$, significantly higher than all other observations.
- **Annual_Visits:** P4 (50 visits) yields $Z = 1.75$, and P8 (40 visits) yields $Z = 1.21$.

ii. Extreme Observations: Records P4 and P8 contain extreme values.

iii. Capping vs. Removal: Capping (Winsorization) is selected over removal to avoid reducing the already small dataset size while neutralizing the influence of severe outliers.

C. Categorical Standardization

i. Inconsistent Gender Values: M, Male, male, F, Female, female, FEMALE

ii. Standardized Set: Male (P1, P2, P7), Female (P3, P4, P5, P6, P8)

iii. Importance: Categorical standardization prevents machine learning models from treating identical categories as distinct variables during encoding, preserving model accuracy.

D. Normalization

i. Min-Max Normalization of BMI [0,1]: Range = [21.8, 60.0]

Normalized values: P1 (0.018), P2 (0.079), P3 (0.202), P4 (0.613), P5 (0.202), P6 (0.246), P7 (0.000), P8 (1.000).

ii. Effect on Performance: Normalization prevents features with large scales from dominant influence in gradient-based and distance-based algorithms.

Question 2: Covariance and Feature Selection in Banking Analytics

A. Covariance Analysis

i. Covariance Calculation: Cov(Income, Loan_Amount) ≈ 2783.39

ii. Relationship: Strong Positive Relationship.

iii. Indication: Indicates that customers with higher income levels systematically apply for and receive higher loan amounts.

B. Feature Relationship Analysis

i. Highly Correlated Pairs: (Income, Loan_Amount), (Income, Monthly_EMI), (Income, Savings), (Loan_Amount, Monthly_EMI)

ii. Redundancy Reason: Higher income enables higher savings, larger loan capacity, and larger monthly EMI payments, introducing severe multicollinearity.

iii. Features to Remove: Loan_Amount, Monthly_EMI, Savings.

C. Feature Selection

i. Optimal Subset: Age, Income, Credit_Score, Risk (Target)

ii. Justification: Credit_Score captures repayment reliability, Income captures capacity, and Age captures stage-of-life demographic risk, eliminating redundancy without loss of core signal.

iii. Cost Reduction: Reducing variables speeds up model convergence and prevents model overfitting.

D. Standardization

i. Z-Score Normalization: Standardized Credit_Score and Income to zero mean and unit variance.

ii. Direct Comparison Explanation: Credit_Score (600–850) and Income (35–130) operate on completely different scales. Unscaled comparison causes optimization algorithms to incorrectly weigh the larger-scale feature.

Question 3: Discretization of Sales Data in Retail Analytics

A. Equal-Width Binning

i & ii. Bin Width Calculation: Width = (6500 - 500) / 4 = 1500

iii. Bin Assignments:

- Bin 1 [500 - 2000]: C1, C2, C3, C4, C5
- Bin 2 (2000 - 3500]: C6, C7
- Bin 3 (3500 - 5000]: C8, C9
- Bin 4 (5000 - 6500]: C10

B. Equal-Frequency Binning

i & ii. Equal-Frequency Assignments:

- Bin 1: C1, C2
- Bin 2: C3, C4, C5
- Bin 3: C6, C7, C8
- Bin 4: C9, C10

iii. Comparison: Equal-width maintains clear metric intervals but creates unbalanced groups. Equal-frequency balances customer volume across bins but distorts interval spacing.

C. Concept Hierarchy

- **Low ($500 - $1000):** C1, C2, C3
- **Medium ($1500 - $2500):** C4, C5, C6
- **High ($3000 - $4000):** C7, C8
- **Very High ($5000 - $6500):** C9, C10

D. Interpretation

i. Improvement: Simplifies continuous transactional numbers into high-level strategic segments for business decisions.

ii. Disadvantage: Results in loss of fine-grained customer variance details within the same category.

Question 4: Data Transformation in Manufacturing Quality Analytics

A. Min-Max Normalization

i & ii. Normalized Values [0,1]: M1 (0.000), M2 (0.077), M3 (0.108), M4 (0.154), M5 (0.231), M6 (0.308), M7 (0.385), M8 (1.000).

iii. Maximum Observation: Machine M8 achieves the maximum normalized score of 1.000.

B. Z-Score Normalization

i. Mean (μ): 73.38°C | **ii. Std Dev (σ):** 20.21°C

iii. Z-Scores: M1 (-0.91), M2 (-0.66), M3 (-0.56), M4 (-0.41), M5 (-0.17), M6 (+0.08), M7 (+0.33), M8 (+2.31).

C. Outlier Analysis

i. Extreme Value: 120°C is an outlier with Z = +2.31 compared to the rest of the operational cluster.

ii. Effect on Mean: Artificially inflates the mean, squishing normal operational machines into a narrow scaled range.

iii. Treatment: Apply Trimming/Removal or Robust Scaling using Median/IQR.

D. Sensitivity Comparison

Method	Sensitivity to Extreme Observations
Min-Max Normalization	Highly Sensitive; extreme values compress normal data ranges.
Z-Score Normalization	Moderately Robust; retains mean centering and unbounded distribution.

Question 5: Missing Value Imputation and Data Transformation in Student Analytics

A. Missing Value Analysis

i. Missing Values Identified: Internal_Mark (S2), Attendance (S3), Study_Hours (S5), Assignment_Score (S6).

ii. Imputations: Study_Hours (Median=5.0, Mean=5.86), Internal_Mark (Median=80.0, Mean=79.43), Attendance (Median=90.0, Mean=86.43), Assignment_Score (Median=78.0, Mean=77.43).

iii. Suitability: Median Imputation (5.0) is better suited for Study_Hours as it resists skewness caused by high values like 12 hours.

B. Data Transformation

i. Min-Max on Internal_Mark: S1 (0.000), S2 (0.568), S3 (0.302), S4 (0.465), S5 (0.581), S6 (0.767), S7 (0.860), S8 (1.000).

ii. Z-Score on Assignment_Score: S1 (-1.52), S2 (-1.09), S3 (-0.47), S4 (+0.05), S5 (+0.40), S6 (0.00), S7 (+1.10), S8 (+1.53).

iii. Comparison: Min-Max bounds scores strictly to [0,1], while Z-Score highlights student performance standard deviations relative to the class mean.

C. Discretization of Study_Hours

- **Low (< 4 hrs):** S1, S2
- **Medium (4 - 7 hrs):** S3, S4, S5, S6
- **High (> 7 hrs):** S7, S8

D. Interpretation

Preprocessing prevents models from giving unfair weight to larger feature scales, improving both prediction accuracy and model interpretability for academic decision-making.